

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
PART TEST – III
PAPER –1
TEST DATE: 24-12-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B, D

Sol. $V = \pi a^2 L$
 $dV = (2\pi a L) da$
 $\Delta V = 2\pi a L (\Delta a)$
 $B = \frac{-\Delta p}{\left(\frac{\Delta V}{V}\right)}, \Delta p = \frac{2B\Delta a}{a}$
 $\Rightarrow F = \mu(\Delta P) \times \text{surface area}$
 $F = \mu \frac{2B\Delta a}{a} (2\pi a L) = 4\pi\mu B L \Delta a$

2. B, D

Sol. Maximum and minimum velocity depends on eccentricity of orbits

$$T.E = -\frac{GmMe}{2a}$$
$$\frac{dA}{dt} = \frac{L}{2m}$$
$$T^2 \propto a^3$$

3. A, D

Sol. $v = \frac{\mu_2}{\mu_1} u = \frac{4/3}{2} (-R)$

$$|v| = \frac{2}{3} R$$

$$\text{Change in magnitude} = \left| \frac{2}{3} R - R \right| = \frac{R}{3}$$

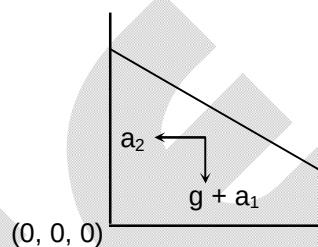
$$\text{Now, } \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{v} - \frac{4/3}{\left(-\frac{5}{3}\right)R} = \left(\frac{1 - \frac{4}{3}}{-R}\right)$$

$$\Rightarrow v = \frac{-15}{7}R$$

4. C

Sol. From the frame of container, the acceleration of a particle is shown in the figure. So we can write the pressure at any point inside the container



5. B

Sol. $P = P_0 \left(\frac{m}{M_s} \right)^{9/2}$

$$-\frac{dm}{dt}c^2 = P_0 \left(\frac{m}{M_s} \right)^{9/2}$$

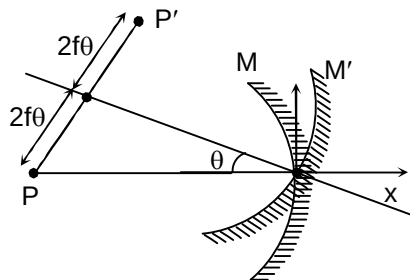
$$\int_{m_0}^m -\frac{dm}{m^{9/2}} = \frac{P_0}{(M_s)^{9/2}c^2} \int_0^t dt$$

$$\frac{2}{7} \left[\frac{1}{m^{7/2}} - \frac{1}{m_0^{7/2}} \right] = \frac{P_0 t}{(M_s)^{9/2}c^2}$$

$$\frac{1}{m^{7/2}} = \frac{1}{m_0^{7/2}} + \frac{7P_0 t}{2(M_s)^{9/2}c^2}$$

6. A

Sol. By rotating an angle ' θ ' the principal axis of mirror also rotated by same angle.



7. D

Sol. Energy density = $\frac{1}{2}\rho A^2\omega^2$

$$0.25\pi^2 = \frac{1}{2}(450)A^2(2\pi f)^2$$

$$\frac{1}{60} = Af$$

...(i)

$$\text{Maximum shear strain} = AK$$

$$A \frac{2\pi}{\lambda} = 3\pi \times 10^{-6}$$

$$\frac{A}{\lambda} = \frac{3}{2} \times 10^{-6} \quad \dots (ii)$$

$$(72) \times \frac{\pi}{180} = \frac{2\pi}{\lambda} \quad (1)$$

$$\lambda = 5\text{m}$$

$$\text{So, } A = 7.5 \times 10^{-6} \text{ m}$$

$$A f = 1/60$$

$$f = \frac{20000}{9}$$

$$\omega = \frac{40000\pi}{9}$$

$$y = \left(\frac{15}{2} \times 10^{-6} \right) \sin \left(\frac{40000}{9} \pi t - \frac{2\pi}{5} x \right)$$

8. Sol.

B

Let $m = 10 \text{ gm}$

$\ell = 20 \text{ cm}$, $k = 200 \text{ N/m}$

$$\text{Angular frequency of } 2m, \omega = \sqrt{\frac{k}{2m}} = 100 \text{ rad/s}$$

$$\text{Angular frequency of } m, \omega' = \sqrt{\frac{2k}{m}} = 2\omega = 200 \text{ rad/s}$$

Let x_1 and x_2 are the displacement of $2m$ and m at the time of oscillation

$$x_1 + x_2 = \ell \quad \dots (i)$$

$$x_1 = \frac{\ell}{2} - \frac{\ell}{2} \cos \omega t \quad \dots (ii)$$

$$x_2 = \frac{\ell}{2} - \frac{\ell}{2} \cos 2\omega t \quad \dots (iii)$$

From (i), (ii) and (iii)

$$\cos \omega t = \frac{1}{2}, \cos 2\omega t = -\frac{1}{2}$$

$$\text{So, } x_2 = \frac{\ell}{2} - \frac{\ell}{2} \left(-\frac{1}{2} \right) = \frac{3\ell}{4}$$

Differentiate equation (ii) and (iii)

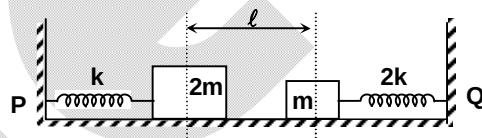
$$v_1 = \frac{dx_1}{dt}, |v_1| = \frac{\sqrt{3}\ell\omega}{4}$$

$$v_2 = \frac{dx_2}{dt}, |v_2| = \frac{\sqrt{3}\ell\omega}{2}$$

Maximum velocity will occur when both springs are at natural length

$$\frac{1}{2}k\left(\frac{\ell}{4}\right)^2 + \frac{1}{2}2k\left(\frac{\ell}{4}\right)^2 = \frac{1}{2}(3m)v_m^2$$

$$\Rightarrow v_m = \frac{\ell}{4} \sqrt{\frac{k}{m}} = 5\sqrt{2}$$



9. C

$$\text{Sol. } \frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

$$\frac{2}{v} - \frac{5/3}{(-12)} = \frac{2 - (3/2)}{(-12)}$$

$$v = \frac{-144}{13} \quad \dots(i)$$

$$\Rightarrow m_T = \frac{v}{u} \times \frac{\mu_1}{\mu_3} = \frac{\left(\frac{-144}{13}\right)}{-12} \times \frac{5/3}{2} = \frac{10}{13} \quad \dots(ii)$$

$$m_L = \left(\frac{v}{u}\right)^2 \frac{\mu_1}{\mu_3} = \left[\frac{\left(\frac{-144}{13}\right)}{-12}\right]^2 \times \frac{5/3}{2} = \frac{120}{169} \quad \dots(iii)$$

$$A_x = (5 \cos 37^\circ) \left(\frac{120}{169}\right) = \frac{480}{169} \text{ mm}$$

$$A_y = (5 \sin 37^\circ) \left(\frac{10}{13}\right) = \frac{30}{13} \text{ mm}$$

$$v_x = (1 \cos 37^\circ) \left(\frac{120}{169}\right) = \frac{96}{169} \text{ m/s}$$

$$v_y = (1 \sin 37^\circ) \left(\frac{10}{13}\right) = \frac{6}{13} \text{ m/s}$$

10. D

$$\text{Sol. For P: } 5 \times 10^{-16} = (1.6 \times 10^{-19}) V_0$$

$$\Rightarrow V_0 = 3125 \text{ volt}$$

$$\text{For Q: } \lambda = \frac{hc}{\Delta E} = \frac{(6.62 \times 10^{-34})(3 \times 10^8)}{35 \times 10^{-18}} = 56.7 \times 10^{-10} \text{ m}$$

$$\text{For R: } v = \frac{E}{h} = \frac{46 \times 10^{-17}}{6.62 \times 10^{-34}} = 6.95 \times 10^{17} \text{ Hz}$$

$$\text{For S: K.E.} = 46 \times 10^{-17} - 5 \times 10^{-18} = 455 \times 10^{-18} \text{ J}$$

11. C

$$\text{Sol. } T = \frac{2\pi r^{3/2}}{\sqrt{GM_e}} = 9.34 \times 10^3 \text{ sec}$$

$$E_i = \frac{-GmM_e}{2r} - \frac{GmM_e}{2r} = \frac{-GmM_e}{r} = 20.84 \times 10^9 \text{ J}$$

$$E_f = -\frac{GmM_e}{r} - \frac{GmM_e}{r} = -\frac{2GmM_e}{r} = 41.68 \times 10^9 \text{ J}$$

$$L = mvr = m\sqrt{GM_e r} = 30.99 \times 10^{12} \text{ kg-m}^2/\text{s}$$

SECTION – B

12. 500

Sol. $v = \sqrt{2gy}$... (i)

Consider a strip at depth 'y'

Volume flow rate from strip

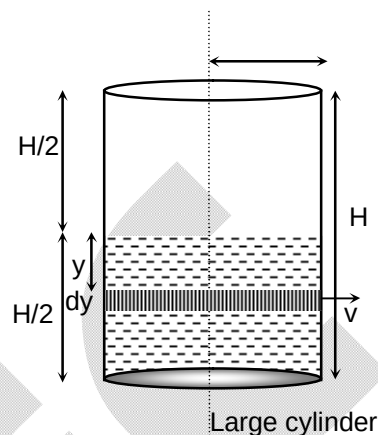
$$dQ = av$$

$$= k(2\pi R)dy\sqrt{2gy}$$

Net volume flow rate

$$Q = k2\pi R\sqrt{2g} \int_0^{H/2} \sqrt{y} dy$$

$$Q = \frac{\pi}{500} R\sqrt{g} H^{3/2}$$



13. 5000

Sol. A central bright is formed at C. I_1 and I_2 are the images formed by M_1 and M_2 is used as point source.For 2nd maxima at P

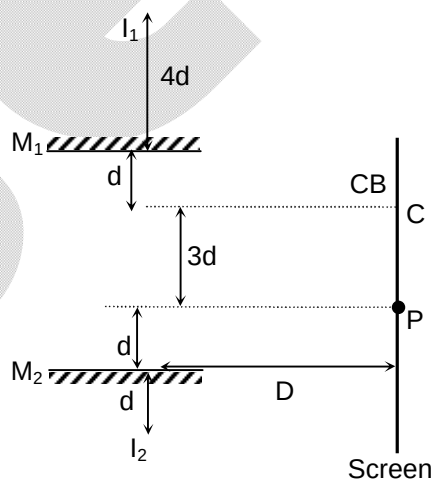
$$\text{Path difference} = 2\lambda$$

$$(d') \sin \theta = 2\lambda$$

$$(10d) \frac{y}{D} = 2\lambda$$

$$\frac{(10d)(3d)}{D} = 2\lambda$$

$$\lambda = \frac{15d^2}{D} = \frac{15(10^{-3})^2}{3} = 5 \times 10^{-6} \text{ m} = 5000 \text{ nm}$$



14. 30

Sol. Let 'x' is the displacement of plank from its equilibrium position

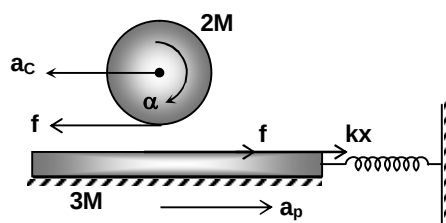
$$f + kx = (3M)a_p \quad \dots (i)$$

$$f.R = \frac{(2M)R^2}{2} \alpha \quad \dots (ii)$$

$$a_c = \frac{f}{2M} \quad \dots (iii)$$

$$a_c + R\alpha = -a_p \quad \dots (iv)$$

$$\text{On solving } \omega = \sqrt{\frac{3k}{11M}} = \sqrt{30} \text{ rad/s}$$



15. 1089

$$\text{Sol. } \frac{5\lambda}{2} = 5, \lambda = 2\text{m}$$

$$\Delta P_0 = 10^3 \text{ N/m}^2$$

$$\Delta P_0 = \rho A K \quad \dots(i)$$

$$\Delta p_0 = \rho A K \quad \dots(ii)$$

$$\frac{\Delta P_0}{\Delta p_0} = \frac{B}{\rho}$$

$$\Delta p_0 = \frac{\rho}{B} \Delta P_0 = \frac{\rho}{\rho v^2} \Delta P_0 \quad \left(v = \sqrt{\frac{B}{\rho}}, B = \rho v^2 \right)$$

$$\Delta p_0 = \frac{\Delta P_0}{v^2} = \frac{1000}{330 \times 330} = \frac{1}{1089}$$

16. 5036

Sol. Pitch = 0.1 mm, N = 100

$$\text{Least count} = \frac{0.1}{100} = 0.001 \text{ mm}$$

$$\text{Negative zero error} = 6 \times 0.001 = 0.006 \text{ mm}$$

$$\text{Reading of screw gauge} = (50 \times 0.1) \text{ mm} + (30 \times 0.001) \text{ mm} = 5.03 \text{ mm}$$

$$\text{Measured value} = (5.03 + 0.006) \text{ mm} = 5.036 \text{ mm}$$

17. 11

Sol. $\frac{12400}{4532} = 2.73 \text{ eV}$

$$\frac{12400}{5112} = 2.42 \text{ eV}$$

$$\frac{12400}{5628} = 2.2 \text{ eV}$$

$$\frac{12400}{6123} = 2.02 \text{ eV}$$

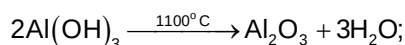
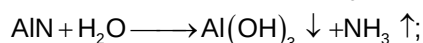
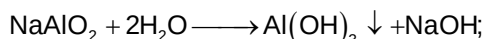
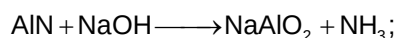
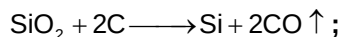
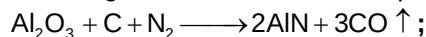
$$\text{No. of emitted photoelectron} = \left(\frac{IA}{4E_1} + \frac{IA}{4E_2} \right) \times t \times \frac{40}{100}$$

$$= \frac{5.6 \times 10^{-4} \times 8 \times 10^{-4}}{4 \times 1.6 \times 10^{-19}} \left[\frac{1}{2.73} + \frac{1}{2.42} \right] \times 3 \times 0.4 = 6.54 \times 10^{11}$$

Chemistry**PART – II****SECTION – A**

18. B, C

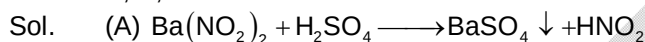
Sol. Following reactions occurs in Serpeck's process:



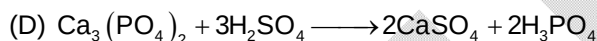
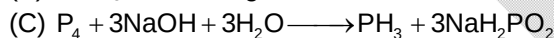
19. B, C

Sol. The given solution is basic buffer.

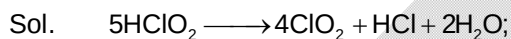
20. A, B, D



(B) HNO_3 is oxidizing acid, which oxidize H_2S to 'S'.



21. D



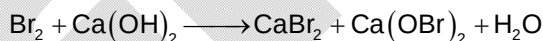
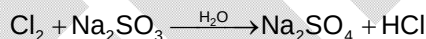
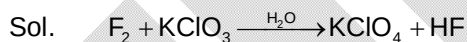
22. B

Sol. $\Lambda_{\text{Na}^+} = 50.11 \text{ cm}^2 / \text{mole}$

$$\Lambda_{\text{Ba}^{2+}} = 127 \text{ Scm}^2 / \text{mole}$$

One Ba^{2+} is replaced by two Na^+ ions.

23. B



24. C

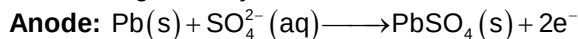
Sol. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{Cl}](\text{NO}_2)\text{Cl}$ are the ionization isomers.

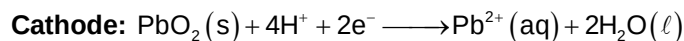
25. C

Sol. Cetyltrimethylammonium bromide is a cationic detergent.
Terfenadine is also known as Seldane.

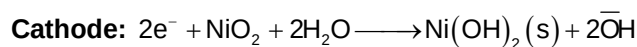
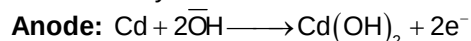
26. C

Sol. Lead storage battery

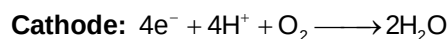
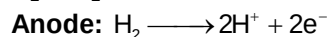




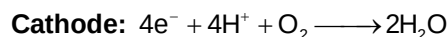
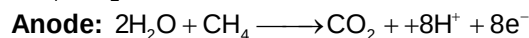
Ni-Cd battery



$\text{H}_2 - \text{O}_2$ fuel cell



$\text{CH}_4 - \text{O}_2$ fuel cell



27.

B

Sol. KO_2 - is a superoxide (O_2^-) which is paramagnetic.

Fe_3O_4 - is Ferrimagnetic.

MnO - is Antiferromagnetic

28.

D

Sol. Cheralite - $\text{CaTh}(\text{PO}_4)_2$

Monazite - (Phosphates & Silicates of - Ce, La, Nd and Th.

Thorite - Silicates of Th and U

Carnotite - $\text{K}_2(\text{UO}_2)_2(\text{VO}_4)_2 \cdot 3\text{H}_2\text{O}$

Samaraskite- (Y) - Oxide of Y, Fe, U, Th, Ca, Nb and Ta

SECTION - B

29. 332



30. 152

Sol. Cr_2O_3 is the green coloured compound which blown in the air.

31. 245

Sol. $M = 0.5$ $N = 1.0$

$$\Lambda_{\text{eq}} = \frac{1000\kappa}{1} \Rightarrow \kappa = 0.98 \text{ unit}$$

$$V = iR = i \times \rho \frac{\ell}{a} = i \times \frac{1}{\kappa} \cdot \frac{\ell}{a}$$

$$25 = i \times \frac{1}{0.98} \times \frac{1}{10}$$

$$i = 245 \text{ (amp)}$$

32. 250

Sol. To prevent 20 ml, amount of 'X' = 2000 mg

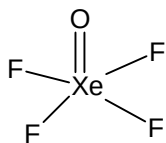
To prevent 10 ml, amount of 'X' = 1000 mg

To prevent from 1 ml, 10% NaCl,

$$\text{Amount of 'X'} = \frac{1000}{2 \times 2} = 250 = \text{Gold number}$$

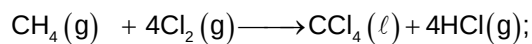
33. 21

Sol.

Number of lone pairs = $1 + 3 \times 4 + 2 = 15$ Number of σ bonds = 5Number of π bonds = 1

34. 2

Sol.



Moles

2 1

reacted at 50% yield

0.125 0.5

Finally

1.875(mole) 0.5(mole) 0.5 mole

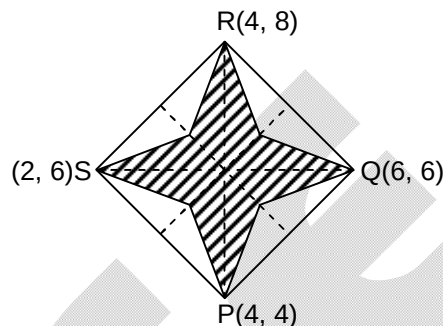
 $PV = nRT$ $P \times 39.6 = 2.875 \times 0.0821 \times 336$ $P = 2 \text{ atm.}$

Mathematics

PART – III

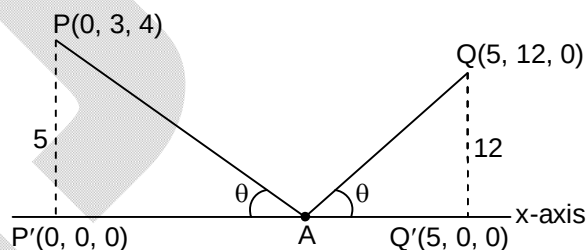
SECTION – A

35. A, C
 Sol. A = Region PQRS
 B = shaded region



36. A, B
 Sol. If $P(z_1)$, $Q(z_2)$, $R(z_3)$ and $S(z_4)$ are points on argand plane
 $\therefore (z_1 - z_2)(z_3 - z_4) + (z_3 - z_1)(z_2 - z_4) + (z_1 - z_4)(z_2 - z_3) = 0$
 $(z_1 - z_2)(z_3 - z_4) = -((z_3 - z_1)(z_2 - z_4) + (z_1 - z_4)(z_2 - z_3))$
 Taking modulus, $(PQ)(RS) \leq (PR)(QS) + (PS)(QR)$ and $(PS)(QR) \leq (PR)(QS) + (PQ)(RS)$

37. A, D
 Sol. $\frac{P'A}{Q'A} = \frac{5}{12}$
 So, $A = \left(\frac{25}{17}, 0, 0\right)$
 Similarly, $B = (0, 7, 0)$
 $C = \left(0, 0, \frac{13}{4}\right)$



38. B
 Sol. For $n = 1$; $\frac{T_1 + 3}{2} = \sqrt{3S_1} \Rightarrow T_1 = 3$
 $(T_n + 3)^2 = 12S_n \Rightarrow (T_{n-1} + 3)^2 = 12S_{n-1}$
 $(T_n + 3)^2 - (T_{n-1} + 3)^2 = 12T_n$
 $T_n - T_{n-1} = 6$ A.P. with common difference = 6

39. C
 Sol. $(U_{n+1} + V_{n+1}i) = (U_n + V_n i)(i + i)$
 $U_1 + V_1 i = \frac{U_{2024} + V_{2024} i}{(1+i)^{2023}}$

40. D
 Sol. By solving equation $x^2 - 2x + 3 = 1$ or 4

41. B

Sol. Let $(2024 - 4r)$ elements have same image and rest $(4r)$ elements is divided into r groups of size 4 $\Rightarrow$ In each group of 4 elements, elements can be arranged in 3 ways

42. C

Sol. Variance = 114 $\Rightarrow \lambda = 6$

$$\text{Mean } (\bar{x}) = \frac{\sum f_i x_i}{\sum f_i} = 24$$

$$\text{Median} = \frac{\left(\frac{N}{2}\right)^{\text{th}} + \left(\frac{N}{2} + 1\right)^{\text{th}} \text{ Obser}}{2} = 24$$

$$\text{Mean Deviation } (\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} = 10$$

$$\text{Mean Deviation } M = \frac{\sum f_i |x_i - M|}{\sum f_i} = 10$$

43. A

Sol. (P) $a = \sum \alpha$; $b = \sum \alpha\beta$; $(-c) = \sum \alpha\beta\gamma$; $d = \sum \alpha\beta\gamma\delta$; $e = \sum \alpha\beta\gamma\delta\lambda$

$$\frac{a + b + (-c) + d}{30} \geq (\alpha^{15} \cdot \beta^{15} \cdot \gamma^{15} \cdot \delta^{15} \cdot \lambda^{15})^{1/30}$$

(Q) $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 19$
 $a + b + c = 19$; $(a - b)^2 + (b - c)^2 + (c - a)^2 = 2$
 Possible values of $|a - b|$ are 0 or 1

(R) $(17 - 4)^{2023} + (17 + 4)^{2023}$ is divisible by 17
 So, will be of the form $51q$, $51q + 17$ or $51q + 34$
 $\therefore 13^{2023} + 21^{2023}$ gives remainder 1 when divided by 3 so, $\lambda = 34$

(S) $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

44. D

Sol. (P) $\lambda = {}^9C_3(4) + {}^9C_2(3) = 444$

(Q) $\frac{a}{b} = \frac{675 \times 675}{2025 \times 2025} = \frac{1}{9}$

(R) ${}^nC_r = {}^{2024}C_3$

(S) Total function = $(2024)^{2023} (1012)^1 = 2^{6071} \cdot 11^{2024} \cdot 23^{2024}$

45. B

Sol. (P) For each matrix having $|A| = k$, there exists a matrix with $|A| = -k$

(Q) $A = B + I$; $B = \begin{bmatrix} 4 & -2 \\ 8 & -4 \end{bmatrix} \Rightarrow B^2 = O$

(R) $f(x) = (x - 1)(x - 3)Q(x) + (2x - 1)$
 $f(0) = 3Q(0) - 1$

(S) $\sum_{r=0}^{2024} r \binom{2024+r}{r} = \frac{4049}{2023} \left(\frac{1}{2025} - \frac{1}{2026} \right)$

SECTION – B

46. 8

Sol. Let $\overline{IP} = \vec{p}$; $\overline{IQ} = \vec{q}$; $\overline{IR} = \vec{r}$

$$2\vec{p} + 3\vec{q} + 5\vec{r} = 0$$

$$A_1 = \frac{1}{2}|\vec{p} \times \vec{q}| ; A_2 = \frac{1}{2}|\vec{q} \times \vec{r}| = \frac{1}{5}|\vec{p} \times \vec{q}| ; A_3 = \frac{1}{2}|\vec{r} \times \vec{p}| = \frac{3}{10}|\vec{p} \times \vec{q}|$$

$$A = \frac{1}{2}|\vec{p} \times \vec{q} + \vec{q} \times \vec{r} + \vec{r} \times \vec{p}| = |\vec{p} \times \vec{q}|$$

47. 22

$$\text{Sol. } \sum_{k=0}^{1012} \left(\frac{{}^{1012}C_k}{{}^{2024}C_k} \sum_{r=k}^{2024} ({}^rC_k \cdot {}^{2024}C_r) \right) = \sum_{k=0}^{1012} \left({}^{1012}C_k \sum_{r=k}^{2024} \frac{{}^rC_k \cdot {}^{2024}C_r}{{}^{2024}C_k} \right)$$

$$\sum_{k=0}^{1012} {}^{1012}C_k \cdot (2)^{2024-k} = 6^{1012} = (7-1)^{1012} = 49q + 22$$

48. 674

Sol. If $(BA)^\mu = B^\gamma A^\mu$, then $(BA)^{\mu+1} = B^{3\gamma+1} A^{\mu+1}$

$$\text{So, } \lambda = \frac{3^{2024} - 1}{2}$$

49. 460

Sol. Doors having perfect square number will have odd number of divisors while doors having not the perfect square number will have even number of divisors.

$\Rightarrow$ Doors number 301 to 600 will have one less divisor

$\Rightarrow$ In doors number 601 to 900, odd number will have one less divisor while even number will have two less divisor

50. 19

Sol. $1^2 + 3^2 + 5^2 - \dots (6n) \text{ terms} = -6^3 n^2$

Here, $n = 169$

51. 72

$$\text{Sol. } \frac{1}{2} \begin{vmatrix} \alpha^2 & \alpha & 1 \\ \beta^2 & \beta & 1 \\ \gamma^2 & \gamma & 1 \end{vmatrix} = 3$$

$$\begin{vmatrix} 4\alpha^2 & (\alpha+\beta)^2 & (\alpha+\gamma)^2 \\ (\alpha+\beta)^2 & 4\beta^2 & (\beta+\gamma)^2 \\ (\alpha+\gamma)^2 & (\beta+\gamma)^2 & 4\gamma^2 \end{vmatrix} = \begin{vmatrix} \alpha^2 & 2\alpha & 1 \\ \beta^2 & 2\beta & 1 \\ \gamma^2 & 2\gamma & 1 \end{vmatrix} \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix} = -72$$